Chapter 1 Test Questions

Vectors, Matrices & Operations

Warmup Test + Interviewer Test — 20 Questions Total

Level 1: Warmup Test (10 Questions)

Covers the foundational definitions and numpy mechanics from Sections 1.1–1.8: vectors, matrices, core operations, matrix multiplication, transpose, and broadcasting.

Section A — Multiple Choice Questions (5)

Q1. What is the shape of the numpy array created by np.array([1, 2, 3])?

A) (1, 3)

B) (3, 1)

C) (3,)

D) (1, 3, 1)

Q2. Which operator performs TRUE matrix multiplication in numpy (not element-wise)?

A) *

B) @

C) np.add()

D) np.multiply()

Q3. If A has shape (4, 6) and B has shape (6, 3), what is the shape of A @ B?

A) (4, 3)

B) (6, 6)

C) (3, 4)

D) Error — shapes don't align

Q4. What is the correct transpose rule for a matrix product $(AB)^T$?

A) $(AB)^T = A^TB^T$

B) $(AB)^T = B^TA^T$

C) $(AB)^T = (AB)^{-1}$

D) $(AB)^T = BA$

Q5. What happens when you compute (1000, 5) - (5,) in numpy?

A) ValueError — shapes incompatible

B) numpy broadcasts the (5,) array across all 1000 rows, result is (1000,5)

C) Only the first 5 rows are affected

D) numpy reshapes to (1000,1000)

Section B — Descriptive Questions (3)

Q6. *[Descriptive]*

Explain the difference between element-wise multiplication (Hadamard product) and matrix multiplication in numpy. Why does mixing them up cause silent bugs?

Answer

Q7. *[Descriptive]*

Why is the dot product described as 'the single most important operation in ML'? Give two concrete examples from real ML workflows.

Answer

Q8. *[Descriptive]*

Explain what feature standardization (Z-score scaling) does mathematically, and why distance-based algorithms like KNN break down without it.

Answer

Section C — Coding Questions (2)

Q9. *[Coding]*

Write numpy code to compute the cosine similarity between vectors a=[1,2,3] and b=[4,5,6] WITHOUT using sklearn. Briefly explain why you can't just use np.dot() alone.

Answer

Q10. *[Coding]*

Given a feature matrix X of shape (100, 5), write code to manually standardize it (Z-score) and explain what would go wrong if you used X.mean() instead of X.mean(axis=0).

Answer

Level 2: Interviewer Test (10 Intermediate / Semi-Advanced Questions)

Applies Chapter 1 concepts to realistic debugging scenarios, model-building tasks, and database-style data — the kind of questions that come up in actual interviews and on-the-job debugging.

Section A — Scenario-Based Multiple Choice Questions (3)

Q1. Scenario: You have a dataset X with shape (200, 10) — 200 samples, 10 features — and a weight vector w with shape (10,). You want predictions for all samples. Which line of code is correct?

A) y_pred = w @ X

B) y_pred = X @ w

C) y_pred = X * w

D) y_pred = w.T @ X.T

Q2. Scenario: A teammate computes a covariance matrix using X @ X instead of X.T @ X, where X is shape (50, 50) — 50 samples, 50 features, same number by coincidence. What actually happens?

A) Raises a ValueError because the shapes can never multiply

B) Silently runs and returns a (50,50) matrix, but it's NOT a real feature-covariance matrix — it's a completely different quantity (sample-to-sample, not feature-to-feature)

C) Returns the exact same answer as X.T @ X for any matrix

D) numpy detects this is a covariance computation and auto-corrects to X.T @ X

Q3. Scenario: Your KNN model's predictions are almost entirely driven by the 'salary' feature (30,000–200,000) and barely affected by 'age' (18–80) or 'years_experience' (0–40). What's the root cause and best fix?

A) KNN is fundamentally a bad algorithm for this data — switch models

B) The Euclidean distance is dominated by the largest-magnitude feature (salary); fix by standardizing all features before computing distances

C) Increase k (number of neighbours) to balance the effect

D) Remove the salary feature entirely since it's causing problems

Section B — Scenario-Based Descriptive Questions (2)

Q4. *[Scenario — Descriptive]*

You're building a 2-layer neural network forward pass manually in numpy. X has shape (64, 20) and you write h = X @ W1 + b1 where W1 is (20, 32) and b1 is (32,) — this works fine. Then for the second layer you write y_pred = h @ W2 + b2 and get a 'shapes not aligned' error. Walk through your debugging process and explain the fix.

Answer

Q5. *[Scenario — Descriptive]*

A junior teammate computed similarity between two product description vectors using just np.dot(a, b) and got a similarity score of 850 for one pair and 12 for another, concluding the first pair is 'way more similar.' You suspect this conclusion might be wrong. Explain what's likely happening and how you'd correct it.

Answer

Section C — Coding Questions (3)

Q6. *[Coding]*

Implement linear regression's closed-form normal equation, including the bias trick (adding a column of ones to X so you don't need a separate intercept term), and use it to predict on new data.

Answer

Q7. *[Coding]*

Write a function that computes the full pairwise cosine similarity matrix for all rows of a matrix X — no Python for-loops, just broadcasting and matrix multiplication.

Answer

Q8. *[Coding]*

Implement a 2-layer neural network forward pass manually (no PyTorch/TensorFlow), given input X and weight/bias parameters for two layers, using ReLU activation on the hidden layer.

Answer

Section D — Database Scenario-Based Questions (2 Coding)

Q9. *[Database Scenario — Coding]*

You have a users table stored as a pandas DataFrame, where each row has a user_id and an embedding column containing a 50-dimensional numpy array representing that user's preferences (learned from a recommender model). Write code to find the top-5 most similar users to a given target user, using cosine similarity.

Answer

Q10. *[Database Scenario — Coding]*

You have an orders table (DataFrame) with columns [user_id, item_id, quantity, price]. Build a user-item interaction matrix and use matrix multiplication to compute total quantity-weighted revenue per user, then per item.

Answer